

A3-PS005-TC504

Explainable BGP Origin and Subprefix Anomaly Monitor

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Problem Description

- BGP has no origin verification, so any AS can announce any prefix.
- **Origin hijack:** the wrong ASN announces someone else's prefix.
- **Subprefix hijack:** a more-specific /24 inside a legitimate /16 wins by longest-prefix match, even when the /16 has a ROA.
- Alerts need clear evidence, and legitimate multi-origin and policy changes must not be flagged.
- **Goal:** replay-based monitor using time-aligned ROAs and an operator prefix inventory, reporting precision, recall and alert latency.

Background

- **BGP:** the origin AS is the last AS in the path, and the longest prefix wins.
- **ROA:** a signed record of (prefix, origin ASN, maxLength).
- **Route Origin Validation (RFC 6811):** Valid, Invalid or NotFound, based on *all covering ROAs*.
- **Limits of ROV:** unsigned prefixes can't be judged, forged origins pass, and loose maxLength allows subprefix attacks.
- **Time-alignment:** old updates checked against today's ROAs give false alerts.
- **Radix tree:** a prefix is a bit string, so covering prefixes are found in at most 32 steps.

Literature & RFC Review

RFC 6811, BGP Prefix Origin Validation [1]
Defines the Valid, Invalid and NotFound states. This is the rule-set of our baseline detector.
RFC 6483, Validation of Route Origination Using RPKI and ROAs [2]
Explains how ROAs are interpreted, including AS0 ROAs and what "Invalid" means.
Morrison (1968) and Sklower (1991), Patricia / Radix Tree [3], [4]
Compressed binary trie for prefix lookup. Sklower's work applied it to routing tables, and we use it to index ROAs with lookups bounded by prefix length.

Proposed Solution

Our proposed solution has pipeline as follows:
MRT feed + ROA snapshot + prefix inventory -> Radix tree -> rule engine -> evidence log -> ML -> metrics.

- **Rule baseline** (using all covering ROAs):

1. ASN and length match: Valid
2. ASN mismatch: Origin Anomaly
3. Length > maxLength: Subprefix Anomaly
4. Otherwise: Unknown

- **Evidence per alert:** prefix, observed origin, expected ASN(s), maxLength, rule fired.
- **ML (after baseline):** Isolation Forest on Unknown routes using origin-based features only.
- **Evaluation:** labelled synthetic anomalies plus legitimate multi-origin and policy-change cases.

Optimization Proposed

	Before	After
ROA store	List	Radix tree
Lookup	O(N)	O(K), K ≤ 32
Matching	Regex	Bitwise ops
Memory	Whole feed in RAM	Streaming
False positives	Static ROAs	Time-aligned ROA

Solution Architecture & Design

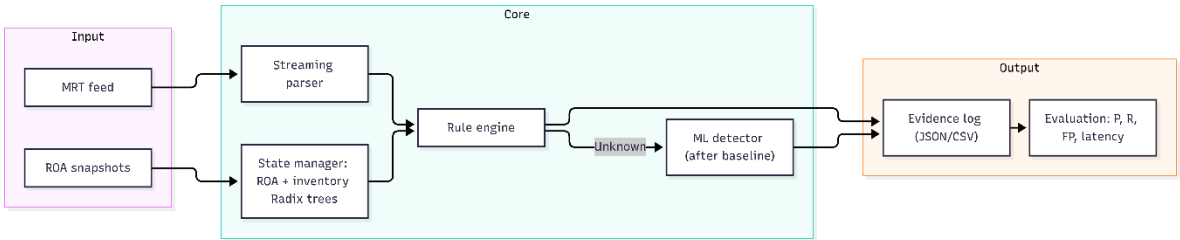


Figure: Architectural Diagram

Data Flow Diagram.

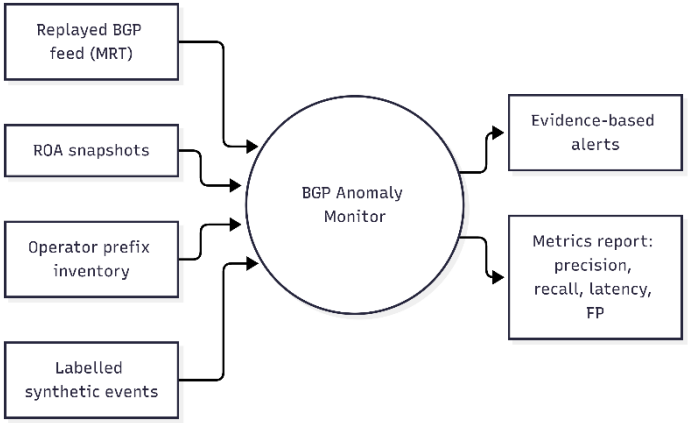


Figure: Level 0 Data Flow Diagram (DFD)

Deployment Timeline

Phase	Date	Target
1	Till 14 Oct	Repo setup, proposal PDF
2	15 to 21 Oct	Parsers and Radix tree
3	22 to 28 Oct	Rule baseline and evidence logs
4	29 Oct to 4 Nov	ML model
5	5 to 10 Nov	False-positive tests, report and demo

Bibliography I

- [1] P. Mohapatra, J. Scudder, D. Ward, R. Bush, and R. Austein, "BGP Prefix Origin Validation," RFC 6811, Internet Engineering Task Force, Jan. 2013. [Online]. Available: https://www.rfc-editor.org/rfc/rfc6811
- [2] G. Huston and G. Michaelson, "Validation of Route Origination Using the Resource Certificate Public Key Infrastructure (PKI) and Route Origin Authorizations (ROAs)," RFC 6483, Internet Engineering Task Force, Feb. 2012. [Online]. Available: https://www.rfc-editor.org/rfc/rfc6483
- [3] D. R. Morrison, "PATRICIA: Practical Algorithm to Retrieve Information Coded in Alphanumeric," *Journal of the ACM*, vol. 15, no. 4, pp. 514–534, Oct. 1968.
- [4] K. Sklower, "A Tree-Based Packet Routing Table for Berkeley Unix," in *Proc. USENIX Winter 1991 Technical Conference*, Dallas, TX, USA, 1991, pp. 93–104.

Thank You for Your Attention!